

WebVM: The Hackathon II!



Problem Statement

Problem:

Many students face difficulties setting up Linux environments and practicing shell commands due to technical barriers and lack of accessible resources.

**Objective:**

Develop a browser-based Linux learning platform that enables seamless access to a Linux environment, allowing users to practice shell commands without complex installations.

Key Features

Real-Time Linux Access:

Embedded Linux terminal powered by **CheerpX** for seamless in-browser virtualization.

Modern Terminal Interface:

Integrated **xterm.js** for a responsive and interactive terminal experience.

Dynamic Learning Modules:

Interactive questions rendered dynamically to guide users through shell command practice.

AI-Powered Assistance:

Built-in **Claude AI** support to answer questions and assist with Bash command execution.

Tech Stack



Frontend:

Built with **React (Next.js)** and styled using **Tailwind CSS** for a responsive and modern user interface.



Terminal Integration:

Leveraging **xterm.js** to provide an interactive and fully functional terminal within the browser.



Virtualization Layer:

Powered by **CheerpX Linux** from Leaning Technologies to run a full Linux environment in-browser without installation.



AI Assistant:

Integrated **Claude 3 Sonnet** via the **Anthropic SDK** for real-time command assistance and question answering.

Demo

Welcome to the Linux Terminal powered by Cheerpx

user@linux-terminal:~\$ |

Explore interactive Linux challenges. Type commands, run scripts, and learn as if you're in your own terminal.

→ List Files

→ Count Words

→ Create a Directory

→ Find a File

→ View File Contents

→ Compress Files

→ Final Challenge: All-in-One Task

Cheerpx

CheerpX: Zero Setup, True Linux in the Browser

- Entire Linux environment runs **entirely on the client side** via **CheerpX**, removing the need for installations or server-side containers.
- Allows **instant, scalable access**.
- Demonstrates the potential of **modern WebAssembly-based virtualization**.

Claude AI:

- Unlike typical AI chat integrations, **Claude** is implemented to **interact directly with the terminal**, offering real-time support with Bash commands and concepts.
- **context-aware**, responding to user inputs and terminal state dynamically.
- Makes Linux more approachable — users can **ask questions in natural language**, get explanations, or even request examples and command execution help.